

Final Project

Machine Learning Applications

Bachelor in Data Science and Engineering

March 23, 2026

1 Introduction

In this project, students will use the knowledge and techniques acquired during the course to solve a machine learning task on text documents. Students can work in groups of a maximum of four people. It is important that, regardless of how each group chooses to distribute the work, all the components of the group know the complete project. The evaluation of the final project will be carried out through the delivery of a report and a group presentation followed by questions.

The project consists of the following tasks:

- **Task 1.** Natural Language Processing and text vectorization
- **Task 2.** Machine Learning:
 - **Task 2.1.** Classification, Regression using feature extraction or selection techniques
 - **Task 2.2.** Clustering using feature extraction or selection techniques
 - **Task 2.3.** Recommendation Systems
- **Task 3.** Final report and presentation.

For the execution of the final project, **students must choose to implement any of the sub-Tasks 2 (either 2.1, 2.2 or 2.3)**, depending on their preferences and the possibilities of the database used.

2 Dataset creation

For the completion of the final project, it is preferred that students work on their own dataset. Said dataset can be a collection of documents available in the open, or it can be created from a collection of Internet pages. Opting for

the second of the proposed alternatives (i.e., creating your own dataset) will positively affect the final grade of the project.

Bear in mind that for the completion of the final project it will be necessary for your dataset to consist of at least several thousand documents in text format, in order to facilitate the construction of sufficiently representative topic models as well as other vectorization approaches. In addition, each document must have additional metadata that can be used for the implementation of Tasks 2.1, 2.2 or 2.3. For instance, to carry out Task 2.3 the dataset needs to contain rating information or in Task 2.1 you need any additional variable (categorical or real) to be used as target of a classification or regression task.

Once you have selected your document collection, **consult with one of the professors** to ensure its viability and discuss potential implementation challenges. We recommend making this selection **as early as possible** to avoid delays that could jeopardize your ability to meet the project deadline.

3 Task 1: Text Preprocessing and vectorization

This task will consist of the thematic analysis of the collection provided. The steps you must follow in your work are as follows:

- **Step 1:** Implementation of a pipeline for the preprocessing of the texts. For this task you could use SpaCy, **and/or** any other library that you consider appropriate.
- **Step 2:** Text vectorization. In this stage, you will analyze the collection by selecting **one method from each of the following four categories**:
 - **Classical approaches:** Either BoW or TF-IDF representation.
 - **Embedding-based approaches:** Either Word2vec, GloVe, or Fast-Text.
 - **Topic modeling-based representation:** Theme extraction and vector representation using the LDA algorithm.
 - **Contextual embeddings:** A Transformer-type model of your choice (e.g., BERT, RoBERTa).

In the report, you must include a description of the preprocessing pipeline used as well as the vectorization strategies that have been explored. For instance, in the embedding-based approaches, you must explain how you convert a set of word vectors into a document vectorization, or for the topic model, you have to explain how you have carried out the selection of the number of topics. Any additional representation that helps to analyze this vectorization will be welcome (and positively evaluated).

4 Task 2: Machine Learning model

For this part of the project, **students must select at least one of the following subtasks.**

Task 2.1: Classification or Regression Task

Implementation and **evaluation** of the performance of a classifier or regression model for the selected dataset. Use one of the metadata available in the dataset as your target variable: a categorical variable if you opt for a classification task, or a real-valued variable for regression. Note that discrete but ordered variables (e.g., dates, scores, etc.) can also be used as target variables for regression.

For this task, you will need to compare the performance by using the different document vectorizations. In addition, **you must use for your work some of the feature extraction or selection algorithms described in the course**, analyzing their impact on the results obtained. For the **performance analysis**, use the metrics appropriate for your task. **For classification**, this includes error rates, ROC curves, and confusion matrices. **If you choose a regression model**, provide the Root Mean Square Error (RMSE) and any other relevant indicators.

To tune the hyperparameters of your classification or regression models, **you must employ a validation methodology and provide a detailed explanation of it in your report.**

Task 2.2: Clustering Task

In case the dataset does not have a clear variable that can be used for document classification or to solve a regression problem, this task can be approached as an unsupervised learning task and document clustering can be performed.

In this task, you **must explore the clustering results produced by the different vectorization schemes from the previous task**. To analyze and compare these results, you should use consistency measures such as the Silhouette Coefficient¹ or perform a qualitative analysis of the clusters (e.g., examining centroids and document distribution).

To determine the optimal number of clusters, you may use the silhouette analysis or other algorithm-specific metrics, such as the Inertia (sum of squared distances to the closest cluster center) for the K-means algorithm. In addition, **you must incorporate feature extraction or selection algorithms discussed in the course**, analyzing their specific impact on the final clustering performance.

Task 2.3: Recommender Systems

In this case, you will have to implement a content based recommendation system using the different document vectorizations obtained from Task 1. You can

¹https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_score.html

easily design a neighborhood-based approach where you recommend to each user similar products to those they already like and use the vectorization of the documents describing each product to calculate these similarities.

The performance of this model must be compared with a collaborative filtering system where you can explore neighborhood based versions (either user based, content based, or both) or latent based methods such as ALS. You can use the Surprise library for the implementation and evaluation of these approaches.

To properly complete this task, **you will have to train the different methods** (selecting their parameters adequately), and evaluate their performance. All these steps have to be clearly explained in the report.

5 Task 3: Final Report and Presentation

For the evaluation of the project, students will need to hand in a written report attaching the code for their implementation. A presentation will also be carried out, where the students must describe their methodology with a few slides.

5.1 Final report

Students must hand in via Aula Global, by the indicated deadline, the following items:

1. Python script or Jupyter notebook with the implemented code duly commented
2. Descriptive report of the work carried out in .pdf format and a maximum length of 12 pages (excluding only cover and references). The report should consist of four main sections:
 - Task 1 (max. 6 pages)
 - Task 2 (max. 5 pages)
 - Acknowledgment of authorship. Inexcusably, the report must respect the principle of recognition of authorship. If you have used extraneous code snippets or any material from external sources, you must clearly specify this in the report. Failing to do so, may result in the loss of the entire grade for the final project.

5.2 Presentation

All team member participants should contribute to the presentation, and individual grades could be granted for this item. Later on, several time slots will be offered, so that groups can book their preferred slot for the presentation.

6 Grading

The maximum mark of the final project is 4 points, which will be distributed as follows:

- Project execution and documentation: 2.5 points.
 - **Task 1.** Natural Language Processing, Topic Modeling and Document Vectorization: **1.25 points**
 - **Task 2.** Machine Classification or Regression/Clustering/Recommendation Systems using characteristics extraction or selection techniques: **1.25 points**

For Tasks 1 and 2 the following aspects will be considered:

- **Methodology (50%):** methodological correctness of your implementation. This includes the correct application of the methods, but also other aspects, such as normalization, hyperparameter validation, selection of evaluation metrics, strategy for evaluating the performance, etc.
 - **Report quality (50%):** the most important aspect that will be assessed is the discussion of your results for which you are encouraged to provide graphical representations supporting your conclusions. Formal presentation will also be taken into account.
- **Project presentation: 1.5 points.** All team member participants should contribute to the presentation, and individual grades could be granted for this item.

Each presentation will be followed by questions by the teachers that can be addressed individually to the different members. The objective of this phase is to verify that all members of each group have participated effectively in the final project, and are aware of its implementation and results in sufficient detail.

If during the Q&A it is found that a member of the team has significant knowledge gaps about the project, the teaching team could apply a penalty factor to that member, which could even result in the total loss of the project's qualification.

Schedule

All Projects must be handed in via Aula Global. The deadline will be Friday, May 10th, at 23:59. Delays will result in a penalty of 0.025 points per hour of delay.

Presentations will take place on May 11th, 12th, and 13th. A doodle will be published for the teams to book time slots for their presentations.